UE24CS352A · Machine Learning mini-project

Keyword Recommendation for Clinical Trial Search

Collaborative filtering on MeSH terms

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Searchers miss relevant keywords

- Short queries omit many medical terms
- BM25 and LSH only match the words the user types
- Goal: predict the MeSH keywords the user forgot, and add them to the query

50,000 trials, 1,000 MeSH terms

50,000

trial records from ClinicalTrials.gov

1,000

most frequent MeSH 2026 terms

80/10/10

random split by trial

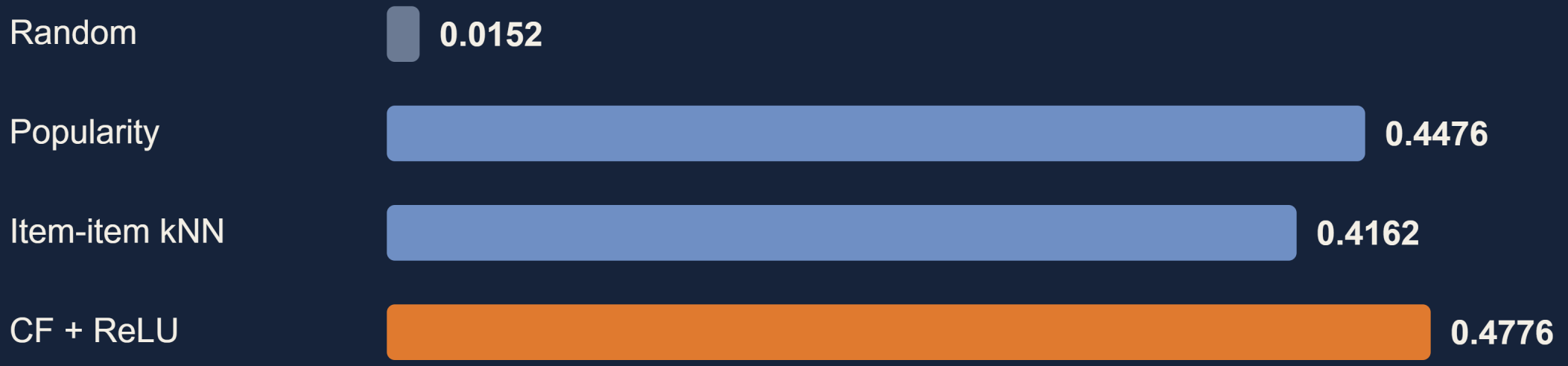
Keywords extracted with a Trie (synonyms mapped to one concept) · random 80/10/10 split

Predict a term from its neighbours

$$\hat{R}_{ij} = ReLU\left(\sum_f W_{if} \cdot \Theta_{if} \cdot R_{fj} + b_i \right)$$

- R: TF-IDF of term i in trial j
- W: fixed cosine-similarity weights, zero diagonal
- Θ and b: learned by mini-batch gradient descent, 60 epochs
- Test: hide 50% of a trial’s terms, rank the top 3

Model comparison: top-3 precision (APTK)



CF + ReLU: about 31× random, +3.0 points over popularity, +6.1 over kNN. Test set, 1,000 MeSH terms.

Live demo

1

Paste a trial description or query

2

See the MeSH keywords found

3

Get the top recommended extra keywords

Conclusions

- Top-3 precision 0.478: about 31× random, +3.0 points over popularity
- +6.1 points over item-kNN; single split, so a modest claim
- Same range as the paper’s 0.50, but not comparable: 1,000 vs 15,071 terms
- Next: more terms, retrieval-quality evaluation